1. Title of the project (5 points):
Production Model Evolution Based Replacement

2. Brief description of the problem the project is expected to address (10 points):
The problem we aim to address is models in production failing over time due to dissatisfaction or overall shifted project goals. Once a model is deployed into production it can serve its role in the product but it may start failing. It might reach a point where its work is just not quite satisfactory or even outright fails. If no downtime can be tolerated a replacement must be found in a timely manner that can replace the current model.

3. Why did your group select this project (5 points):
This project was picked due to an interest in new models being deployed automatically & rolled back to maintain a constant improvement over a longer period of time.

4. The objective of your group (10 points)
When a model fails we want to introduce a survival of the fittest evolutionary system where the model will be mutated & deployed tested in a benchmark if its worse it will be rolled back if its better it will replace the production model. We plan to implement this system with a proof of concept version where the “production” model will be playing a game & we can use the game as a scoring benchmark to determine failure points & improvements over time.

Name	Date	Present	Assigned
Zach Lavoie	9/10/2026	yes	Documentation work
Avery Mackey	9/10/2026	yes	Documentation work